

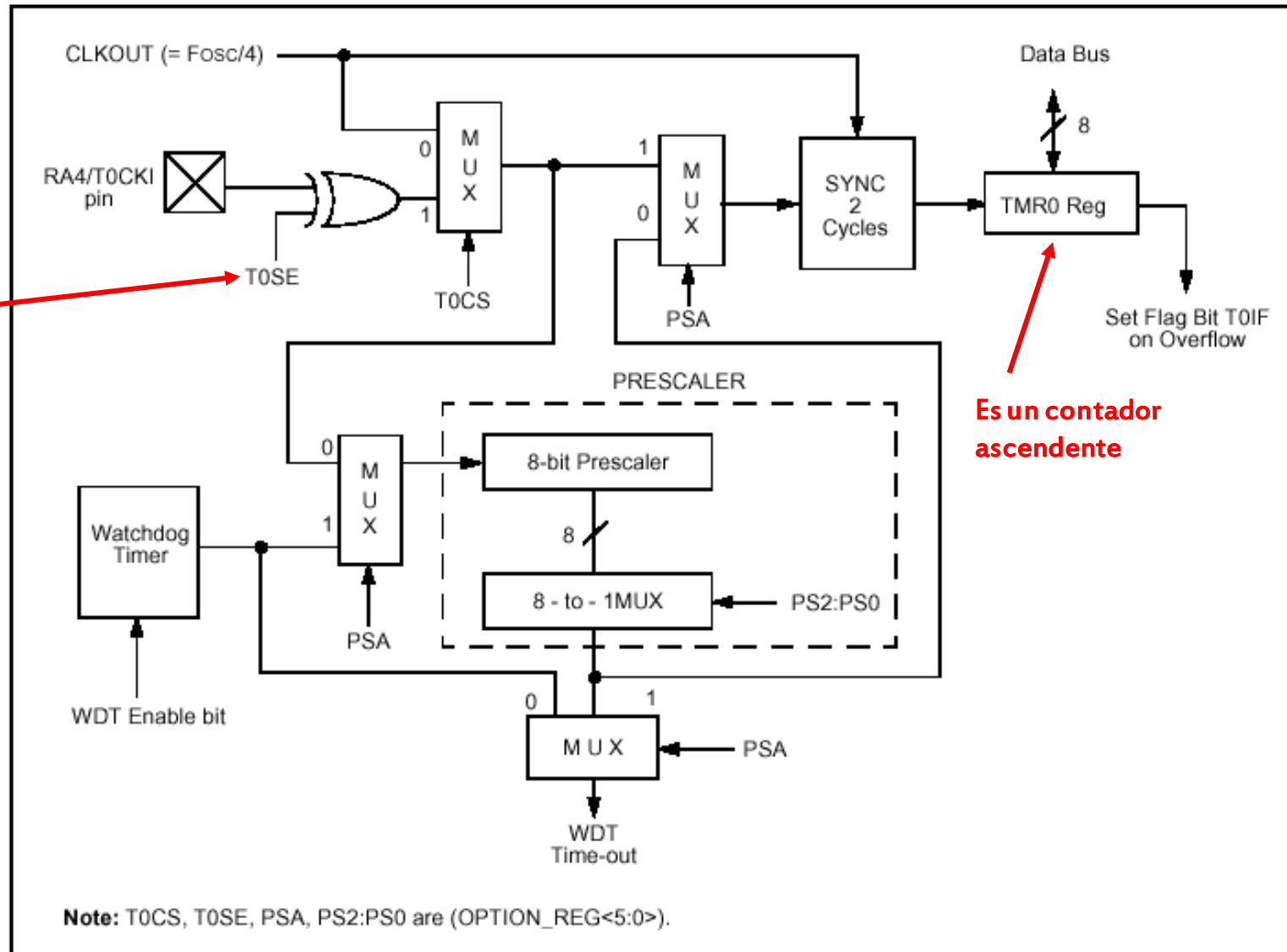
PIC16F876: TEMPORIZADORES: TIMER0

Si $F_{osc} = 4 \text{ MHz}$ ($T = 250 \text{ nS}$) tenemos $CLKOUT = 250 \text{ KHz}$ ($1 \mu\text{S}$)

The Timer0 module timer/counter has the following features:

- 8-bit timer/counter
- Readable and writable
- 8-bit software programmable prescaler
- Internal or external clock select
- Interrupt on overflow from FFh to 00h
- Edge select for external clock

FIGURE 5-1: BLOCK DIAGRAM OF THE TIMER0/WDT PRESCALER



TOSE=1

TOSE=0

NOTA:

Si TOSE = 1
es un inversor

Si TOSE = 0
no invierte

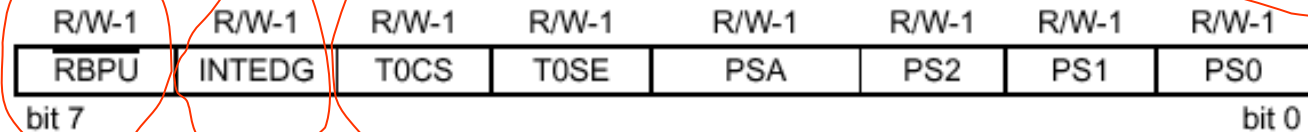
Es un contador
ascendente

PIC16F876: TEMPORIZADORES: TIMER0

“ 6 bits para controlar el temporizador TMR0 y el temporizador de vigilancia WDT, 1 bit para configurar las interrupciones producidas por los flancos en el pin RB0/INT y 1 bit para habilitar las resistencias de “PULL-UP” en el puerto B”

REGISTER 5-1: OPTION REG REGISTER

Resistencias de
PULL-UP del
PORT B



- bit 7 **RBPU**
- bit 6 **INTEDG**
- bit 5 **T0CS**: TMR0 Clock Source Select bit
1 = Transition on T0CKI pin
0 = Internal instruction cycle clock (CLKOUT)
- bit 4 **T0SE**: TMR0 Source Edge Select bit
1 = Increment on high-to-low transition on T0CKI pin
0 = Increment on low-to-high transition on T0CKI pin
- bit 3 **PSA**: Prescaler Assignment bit
1 = Prescaler is assigned to the WDT
0 = Prescaler is assigned to the Timer0 module
- bit 2-0 **PS2:PS0**: Prescaler Rate Select bits

Bit Value	TMR0 Rate	WDT Rate
000	1 : 2	1 : 1
001	1 : 4	1 : 2
010	1 : 8	1 : 4
011	1 : 16	1 : 8
100	1 : 32	1 : 16
101	1 : 64	1 : 32
110	1 : 128	1 : 64
111	1 : 256	1 : 128

Nota:

Lo máximo que se
puede temporizar
vale $256 \times 256 \mu\text{s} =$
 $65,5 \text{ mS}$

$T_{\text{RELOJ}} = 256 \mu\text{s}$

Interrupción en RB0

INTEDG=0



INTEDG=1



PIC16F876: TEMPORIZADORES: TIMER0

Proceso para utilizar el TIMER0 como temporizador.

- 1.- Configurar adecuadamente**
- 2.- Cargar TMR0 con el valor adecuado**
- 3.- Esperar a que se ponga a 1 el bit TOIF (bit 2 de INTCON)**
- 4.- Borrar este bit**

Registro de interrupciones: INTCON

“Habilitar las distintas interrupciones y identificar la fuente que las ha producido”

R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-x
GIE	PEIE	TOIE	INTE	RBIE	TOIF	INTF	RBIF
bit7			bit0				

R = Readable bit
W = Writable bit
U = Unimplemented bit, read as '0'
- n = Value at POR reset
x = unknown

bit 7: **GIE:**⁽¹⁾ Global Interrupt Enable bit
1 = Enables all un-masked interrupts
0 = Disables all interrupts

bit 6: **PEIE:**⁽²⁾ Peripheral Interrupt Enable bit
1 = Enables all un-masked peripheral interrupts
0 = Disables all peripheral interrupts

bit 5: **TOIE:** TMR0 Overflow Interrupt Enable bit
1 = Enables the TMR0 overflow interrupt
0 = Disables the TMR0 overflow interrupt

bit 4: **INTE:** RB0/INT External Interrupt Enable bit
1 = Enables the RB0/INT external interrupt
0 = Disables the RB0/INT external interrupt

bit 3: **RBIE:** RB Port Change Interrupt Enable bit
1 = Enables the RB port change interrupt
0 = Disables the RB port change interrupt

bit 2: **TOIF:** TMR0 Overflow Interrupt Flag bit
1 = TMR0 register overflowed (must be cleared in software)
0 = TMR0 register did not overflow

bit 1: **INTF:** RB0/INT External Interrupt Flag bit
1 = The RB0/INT external interrupt occurred (must be cleared in software)
0 = The RB0/INT external interrupt did not occur

bit 0: **RBIF:** RB Port Change Interrupt Flag bit
1 = At least one of the RB7:RB4 pins changed state (see Section 5.2 to clear the interrupt)
0 = None of the RB7:RB4 pins have changed state

Note 1: For the PIC16C61/62/64/65, if an interrupt occurs while the GIE bit is being cleared, the GIE bit may unintentionally be re-enabled by the RETFIE instruction in the user's Interrupt Service Routine. Refer to Section 13.5 for a detailed description.

2: The PEIE bit (bit6) is unimplemented on the PIC16C61, read as '0'.

Interrupt flag bits get set when an interrupt condition occurs regardless of the state of its corresponding enable bit or the global enable bit, GIE (INTCON<7>). User software should ensure the appropriate interrupt flag bits are clear prior to enabling an interrupt.

PIC16F876:
TEMPORIZADORES: TIMER0

TABLE 5-1: REGISTERS ASSOCIATED WITH TIMER0

Address	Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Value on: POR, BOR	Value on all other RESETS
01h,101h	TMR0	Timer0 Module's Register								xxxx xxxx	uuuu uuuu
0Bh,8Bh, 10Bh,18Bh	INTCON	GIE	PEIE	TOIE	INTE	RBIE	TOIF	INTF	RBIF	0000 000x	0000 000u
81h,181h	OPTION_REG	RBPV	INTEDG	T0CS	T0SE	PSA	PS2	PS1	PS0	1111 1111	1111 1111

PIC16F876: TEMPORIZADORES: TIMER0

UNA PARTE DE INTCON Y DE OPTION
ESTA LIGADA CON LA GESTIÓN DEL
TEMPORIZADOR TIMER0

FIGURE 2-3: PIC16F877/876 REGISTER FILE MAP

File Address	File Address	File Address	File Address
Indirect addr. ⁽¹⁾ 00h	Indirect addr. ⁽¹⁾ 80h	Indirect addr. ⁽¹⁾ 100h	Indirect addr. ⁽¹⁾ 180h
TMR0 01h	OPTION_REG 81h	TMR0 101h	OPTION_REG 181h
PCL 02h	PCL 82h	PCL 102h	PCL 182h
STATUS 03h	STATUS 83h	STATUS 103h	STATUS 183h
FSR 04h	FSR 84h	FSR 104h	FSR 184h
PORTA 05h	TRISA 85h		
PORTB 06h	TRISB 86h	PORTB 106h	TRISB 186h
PORTC 07h	TRISC 87h		
PORTD 08h	TRISD 88h		
PORTEN 09h	TRISE 89h		
PCLATH 0Ah	PCLATH 8Ah	PCLATH 10Ah	PCLATH 18Ah
INTCON 0Bh	INTCON 8Bh	INTCON 10Bh	INTCON 18Bh
PIR1 0Ch	PIE1 8Ch	EEDATA 10Ch	EECON1 18Ch
PIR2 0Dh	PIE2 8Dh	EEADR 10Dh	EECON2 18Dh
TMR1L 0Eh	PCON 8Eh	EEDATH 10Eh	Reserved ² 18Eh
TMR1H 0Fh		EEADRH 10Fh	Reserved ² 18Fh
T1CON 10h			
TMR2 11h	SSPCON2 91h		
T2CON 12h	PR2 92h		
SSPBUF 13h	SSPAD0 93h		
SSPCON 14h	SSPSTAT 94h		
CCPR1L 15h			
CCPR1H 16h			
CCP1CON 17h			
RCSTA 18h	TXSTA 98h		
TXREG 19h	SPBRG 99h		
RCREG 1Ah			
CCPR2L 1Bh			
CCPR2H 1Ch			
CCP2CON 1Dh			
ADRESH 1Eh	ADRESL 9Eh		
ADCON0 1Fh	ADCON1 9Fh		
General Purpose Register 96 Bytes 20h-7Fh	General Purpose Register 80 Bytes A0h-EFh	General Purpose Register 80 Bytes 100h-16Fh	General Purpose Register 80 Bytes 180h-1EFh
	accesses 70h-7Fh	accesses 70h-7Fh	accesses 70h-7Fh
Bank 0	Bank 1	Bank 2	Bank 3

Unimplemented data memory locations, read as '0'.
 * Not a physical register.

Note 1: These registers are not implemented on the PIC16F876.
Note 2: These registers are reserved, maintain these registers clear.

PIC16F876:

TEMPORIZADORES:TIMER1

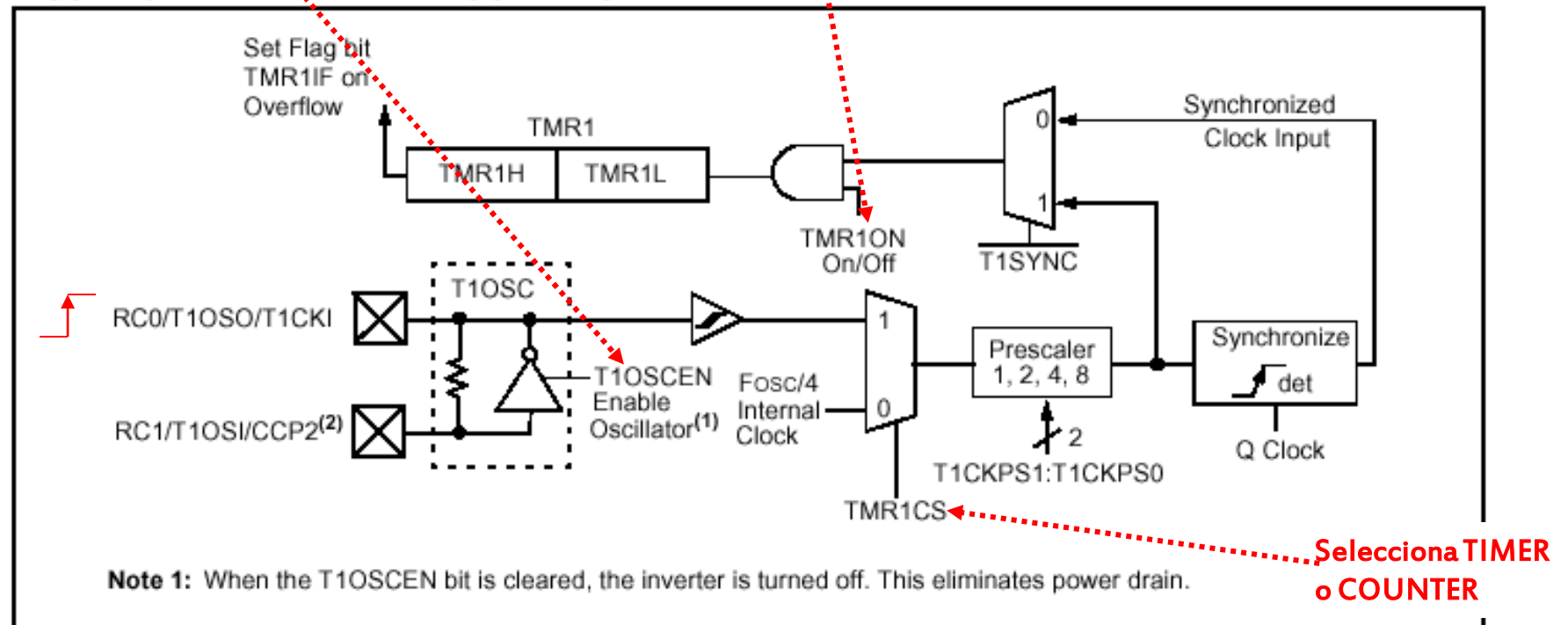
Activa oscilador externo (con 1)

En modo TIMER incrementa cada ciclo de instrucción ($F_{osc}/4$).

En modo COUNTER incrementa cada flanco de subida externo.

Desactivación.
Por defecto está desactivado

FIGURE 6-2: TIMER1 BLOCK DIAGRAM



Como TIMER: El máximo tiempo es: $2^{16} \times 8 \times 1 \mu S = 0.52 S$

PIC16F876: TEMPORIZADORES: TIMER1

REGISTER 6-1: T1CON: TIMER1 CONTROL REGISTER (ADDRESS 10h)

U-0	U-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0
—	—	T1CKPS1	T1CKPS0	T1OSCEN	T1SYNC	TMR1CS	TMR1ON
bit 7							bit 0

bit 7-6 **Unimplemented:** Read as '0'

bit 5-4 **T1CKPS1:T1CKPS0:** Timer1 Input Clock Prescale Select bits

11 = 1:8 Prescale value

10 = 1:4 Prescale value

01 = 1:2 Prescale value

00 = 1:1 Prescale value

bit 3 **T1OSCEN:** Timer1 Oscillator Enable Control bit

1 = Oscillator is enabled

0 = Oscillator is shut-off (the oscillator inverter is turned off to eliminate power drain)

bit 2 **T1SYNC:** Timer1 External Clock Input Synchronization Control bit

When TMR1CS = 1:

1 = Do not synchronize external clock input

0 = Synchronize external clock input

When TMR1CS = 0:

This bit is ignored. Timer1 uses the internal clock when TMR1CS = 0.

bit 1 **TMR1CS:** Timer1 Clock Source Select bit

1 = External clock from pin RC0/T1OSO/T1CKI (on the rising edge)

0 = Internal clock (FOSC/4)

bit 0 **TMR1ON:** Timer1 On bit

1 = Enables Timer1

0 = Stops Timer1

PIC16F876:
TEMPORIZADORES: TIMER1

TABLE 6-2: REGISTERS ASSOCIATED WITH TIMER1 AS A TIMER/COUNTER

Address	Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Value on: POR, BOR	Value on all other RESETS
0Bh, 8Bh, 10Bh, 18Bh	INTCON	GIE	PEIE	T0IE	INTE	RBIE	T0IF	INTF	RBIF	0000 000x	0000 000u
0Ch	PIR1	PSPIF ⁽¹⁾	ADIF	RCIF	TXIF	SSPIF	CCP1IF	TMR2IF	TMR1IF	0000 0000	0000 0000
8Ch	PIE1	PSPIE ⁽¹⁾	ADIE	RCIE	TXIE	SSPIE	CCP1IE	TMR2IE	TMR1IE	0000 0000	0000 0000
0Eh	TMR1L	Holding Register for the Least Significant Byte of the 16-bit TMR1 Register								xxxx xxxx	uuuu uuuu
0Fh	TMR1H	Holding Register for the Most Significant Byte of the 16-bit TMR1 Register								xxxx xxxx	uuuu uuuu
10h	T1CON	—	—	T1CKPS1	T1CKPS0	T1OSCEN	T1SYNC	TMR1CS	TMR1ON	--00 0000	--uu uuuu

Legend: x = unknown, u = unchanged, - = unimplemented, read as '0'. Shaded cells are not used by the Timer1 module.

Note 1: Bits PSPIE and PSPIF are reserved on the PIC16F873/876; always maintain these bits clear.

PIC16F876: TEMPORIZADORES: TIMER1

UNA PARTE DE INTCON, PIR1 Y PIE1
ESTA LIGADA CON LA GESTIÓN DEL
TEMPORIZADOR TIMER1

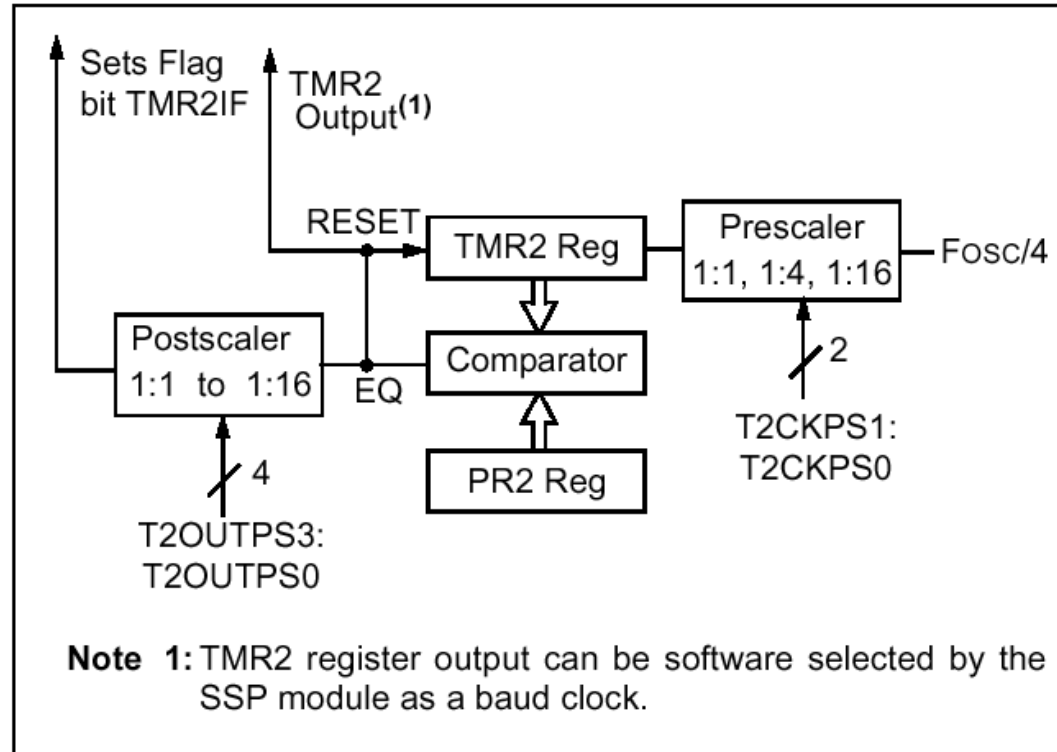
FIGURE 2-3: PIC16F877/876 REGISTER FILE MAP

File Address	File Address	File Address	File Address
Indirect addr. ⁽¹⁾ 00h	Indirect addr. ⁽¹⁾ 80h	Indirect addr. ⁽¹⁾ 100h	Indirect addr. ⁽¹⁾ 180h
TMR0 01h	OPTION_REG 81h	TMR0 101h	OPTION_REG 181h
PCL 02h	PCL 82h	PCL 102h	PCL 182h
STATUS 03h	STATUS 83h	STATUS 103h	STATUS 183h
FSR 04h	FSR 84h	FSR 104h	FSR 184h
PORTA 05h	TRISA 85h	PORTB 105h	TRISA 185h
PORTB 06h	TRISB 86h	PORTB 106h	TRISB 186h
PORTC 07h	TRISC 87h	PORTB 107h	TRISB 187h
PORTD 08h	TRISD 88h		188h
PORTEN 09h	TRISE 89h		189h
PCLATH 0Ah	PCLATH 8Ah	PCLATH 10Ah	PCLATH 18Ah
INTCON 0Bh	INTCON 8Bh	INTCON 10Bh	INTCON 18Bh
PIR1 0Ch	PIE1 8Ch	EEDATA 10Ch	EECON1 18Ch
PIR2 0Dh	PIE2 8Dh	EEADR 10Dh	EECON2 18Dh
TMR1L 0Eh	PCON 8Eh	EEDATH 10Eh	Reserved ² 18Eh
TMR1H 0Fh		EEADRH 10Fh	Reserved ² 18Fh
T1CON 10h			190h
TMR2 11h	SSPCON2 91h		191h
T2CON 12h	PR2 92h		192h
SSPBUF 13h	SSPAD0 93h		193h
SSPCON 14h	SSPSTAT 94h		194h
CCPR1L 15h			195h
CCPR1H 16h			196h
CCP1CON 17h			197h
RCSTA 18h	TXSTA 98h		198h
TXREG 19h	SPBRG 99h		199h
RCREG 1Ah			19Ah
CCPR2L 1Bh			19Bh
CCPR2H 1Ch			19Ch
CCP2CON 1Dh			19Dh
ADRESH 1Eh	ADRESL 9Eh		19Eh
ADCON0 1Fh	ADCON1 9Fh		19Fh
			1A0h
General Purpose Register 96 Bytes	General Purpose Register 80 Bytes	General Purpose Register 80 Bytes	General Purpose Register 80 Bytes
	accesses 70h-7Fh	accesses 70h-7Fh	accesses 70h-7Fh
Bank 0 7Fh	Bank 1 FFh	Bank 2 17Fh	Bank 3 1FFh

 Unimplemented data memory locations, read as '0'.
 * Not a physical register.

Note 1: These registers are not implemented on the PIC16F876.
Note 2: These registers are reserved, maintain these registers clear.

FIGURE 7-1: TIMER2 BLOCK DIAGRAM



Cuenta hasta que iguala PR2 y luego se reinicia

El tiempo máximo es:

$$1 \mu\text{S} \times 16 \times 256 \times 16 = 65.5 \text{ ms}$$

The prescaler and postscaler counters are cleared when any of the following occurs:

- a write to the TMR2 register
- a write to the T2CON register
- any device RESET (POR, $\overline{\text{MCLR}}$ Reset, WDT Reset, or BOR)

TMR2 is not cleared when T2CON is written.

PIC16F876: TEMPORIZADORES: TIMER2

REGISTER 7-1: T2CON: TIMER2 CONTROL REGISTER (ADDRESS 12h)

U-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0
—	TOUTPS3	TOUTPS2	TOUTPS1	TOUTPS0	TMR2ON	T2CKPS1	T2CKPS0
bit 7							bit 0

bit 7 **Unimplemented:** Read as '0'

bit 6-3 **TOUTPS3:TOUTPS0:** Timer2 Output Postscale Select bits

0000 = 1:1 Postscale

0001 = 1:2 Postscale

0010 = 1:3 Postscale

•

•

•

1111 = 1:16 Postscale

bit 2 **TMR2ON:** Timer2 On bit

1 = Timer2 is on

0 = Timer2 is off

bit 1-0 **T2CKPS1:T2CKPS0:** Timer2 Clock Prescale Select bits

00 = Prescaler is 1

01 = Prescaler is 4

1x = Prescaler is 16

PIC16F876:
TEMPORIZADORES: TIMER2

TABLE 7-1: REGISTERS ASSOCIATED WITH TIMER2 AS A TIMER/COUNTER

Address	Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Value on: POR, BOR	Value on all other RESETS
0Bh,8Bh, 10Bh,18Bh	INTCON	GIE	PEIE	T0IE	INTE	RBIE	T0IF	INTF	RBIF	0000 000x	0000 000u
0Ch	PIR1	PSPIF ⁽¹⁾	ADIF	RCIF	TXIF	SSPIF	CCP1IF	TMR2IF	TMR1IF	0000 0000	0000 0000
8Ch	PIE1	PSPIE ⁽¹⁾	ADIE	RCIE	TXIE	SSPIE	CCP1IE	TMR2IE	TMR1IE	0000 0000	0000 0000
11h	TMR2	Timer2 Module's Register								0000 0000	0000 0000
12h	T2CON	—	TOUTPS3	TOUTPS2	TOUTPS1	TOUTPS0	TMR2ON	T2CKPS1	T2CKPS0	-000 0000	-000 0000
92h	PR2	Timer2 Period Register								1111 1111	1111 1111

PIC16F876: TEMPORIZADORES: TIMER2

UNA PARTE DE INTCON, PIR1 Y PIE1
ESTA LIGADA CON LA GESTIÓN DEL
TEMPORIZADOR TIMER2

FIGURE 2-3: PIC16F877/876 REGISTER FILE MAP

File Address	File Address	File Address	File Address
Indirect addr. ⁽¹⁾ 00h	Indirect addr. ⁽¹⁾ 80h	Indirect addr. ⁽¹⁾ 100h	Indirect addr. ⁽¹⁾ 180h
TMR0 01h	OPTION_REG 81h	TMR0 101h	OPTION_REG 181h
PCL 02h	PCL 82h	PCL 102h	PCL 182h
STATUS 03h	STATUS 83h	STATUS 103h	STATUS 183h
FSR 04h	FSR 84h	FSR 104h	FSR 184h
PORTA 05h	TRISA 85h	PORTB 105h	TRISA 185h
PORTB 06h	TRISB 86h	PORTB 106h	TRISB 186h
PORTC 07h	TRISC 87h	PORTB 107h	TRISB 187h
PORTD 08h	TRISD 88h		188h
PORTEN 09h	TRISE 89h		189h
PCLATH 0Ah	PCLATH 8Ah	PCLATH 10Ah	PCLATH 18Ah
INTCON 0Bh	INTCON 8Bh	INTCON 10Bh	INTCON 18Bh
PIR1 0Ch	PIE1 8Ch	EEDATA 10Ch	EECON1 18Ch
PIR2 0Dh	PIE2 8Dh	EEADR 10Dh	EECON2 18Dh
TMR1L 0Eh	PCON 8Eh	EEDATH 10Eh	Reserved ⁽²⁾ 18Eh
TMR1H 0Fh		EEADRH 10Fh	Reserved ⁽²⁾ 18Fh
T1CON 10h			190h
TMR2 11h	SSPCON2 91h		191h
T2CON 12h	PR2 92h		192h
SSPBUF 13h	SSPAD0 93h		193h
SSPCON 14h	SSPSTAT 94h		194h
CCPR1L 15h			195h
CCPR1H 16h			196h
CCP1CON 17h			197h
RCSTA 18h	TXSTA 98h		198h
TXREG 19h	SPBRG 99h		199h
RCREG 1Ah			19Ah
CCPR2L 1Bh			19Bh
CCPR2H 1Ch			19Ch
CCP2CON 1Dh			19Dh
ADRESH 1Eh	ADRESL 9Eh		19Eh
ADCON0 1Fh	ADCON1 9Fh		19Fh
			1A0h
General Purpose Register 96 Bytes	General Purpose Register 80 Bytes	General Purpose Register 80 Bytes	General Purpose Register 80 Bytes
Bank 0 7Fh	Bank 1 FFh	Bank 2 17Fh	Bank 3 1FFh
	accesses 70h-7Fh	accesses 70h-7Fh	accesses 70h-7Fh

 Unimplemented data memory locations, read as '0'.
 * Not a physical register.

Note 1: These registers are not implemented on the PIC16F876.
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